Written Exam term 4 – 2015

Year 9

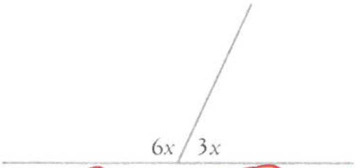
Name: SOLUTIONS

Mark /99

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

1 Find the values of the angles and/or pronumerals in each diagram. GIVE REASONS

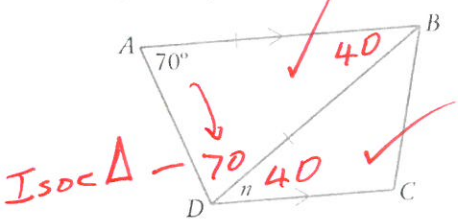
a (3 marks) Find x.



$9x = 180$ ✓ SUPP ∠'s ✓

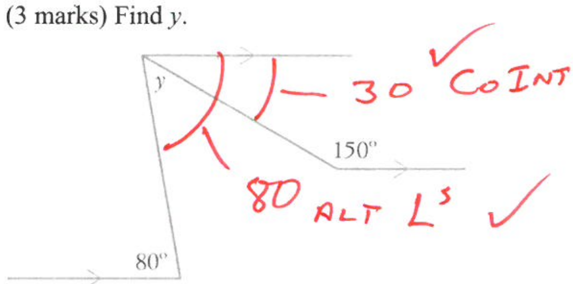
$x = 20°$ ✓

b (3 marks) Find n.



$n = 40°$ ALT ∠'s ✓

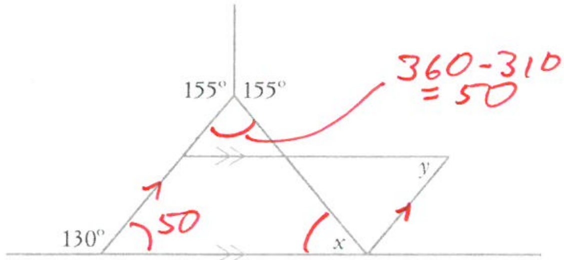
c (3 marks) Find y.



$y = 80 - 30$ ✓

$= 50°$

d (5 marks) Find x and y.



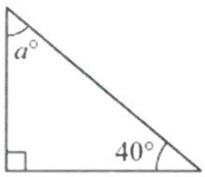
$x = 180 - (50+50)$

$= 80°$

$y = 50°$ OP ∠'s PARALLOG

2 Find the value of each pronumeral.

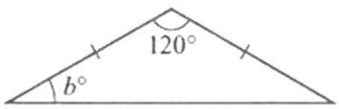
a



$a = 90 - 40$ ✓

$= 50°$ ✓

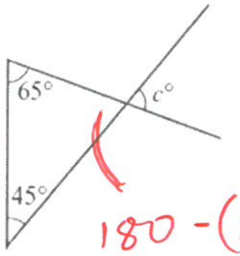
b



$2b = 180 - 120$ ✓

$b = 30°$ ✓

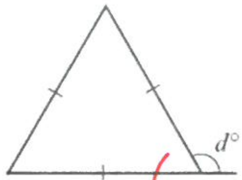
c



$180-(65+45)$
$= 70$

$c = 70°$

d



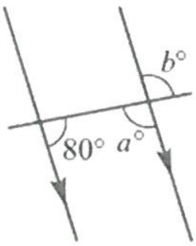
$60°$

$d = 180 - 60$
$= 120°$

(4 × 2 = 8 marks)

3 Find the value of each of the pronumerals. Give reasons for your answers.

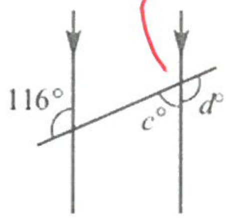
a



$a = 100$ CO INT
$b = 100$ VERT OP

b

$116°$ CORR ∠'s

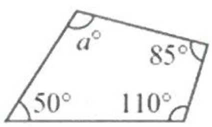


$c = 64°$ SUPP ∠'s
$d = 116$ VERT OPP ∠s

(3 + 3 = 6 marks)

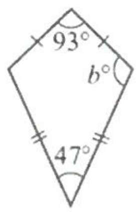
4 Find the value of the pronumeral in each of these polygons.

a



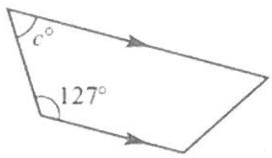
$a = 360-(50+85+110)$
$= 115°$

b



$2b = 360-(93+47)$
$= 220°$
$b = 110°$

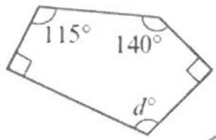
c



$c = 180 - 127^\circ$

$= 53^\circ$ ✓

d

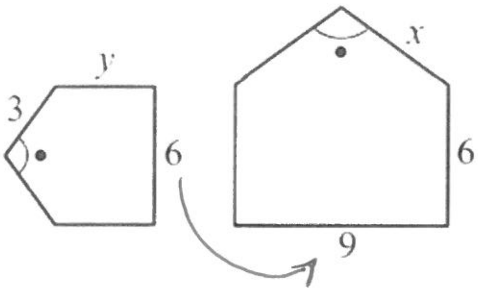


$d = 540 - (90+90+140+115)$

$= 105^\circ$ ✓

(3 × 2 + 3 = 9 marks)

5 This pair of figures are similar.



a Find a scale factor. $\times 1.5$ ✓✓

1.5 or $\frac{3}{2}$

b Find the value of *x*.

$x = 3 \times 1.5 = 4.5$ ✓ ✓

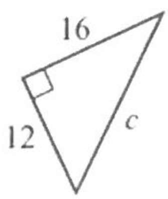
c Find the value of *y*.

$y = 6 \div 1.5 = 4$ ✓ ✓

(2 + 2 + 2 = 6 marks)

6 Find the length of the hypotenuse in each of the following right-angled triangles, leaving your answer to part **b** ~~as an exact value~~. To 2dp

a

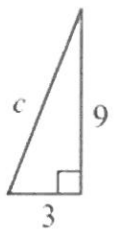


$C = \sqrt{12^2 + 16^2}$ ✓

$= \sqrt{400}$

$C = 20$ ✓

b



$C = \sqrt{3^2 + 9^2}$ ✓

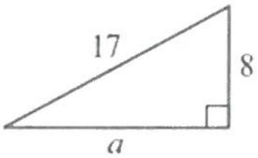
$= \sqrt{90}$

$C = 9.49$ ✓

(2 + 2 = 4 marks)

7 Find the value of the pronumeral in each of the following right-angled triangle, leaving your answer to part **b** ~~as an exact value.~~ To 2dp

a

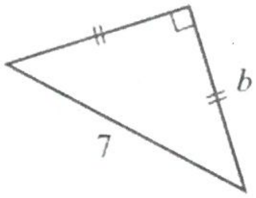


$a = \sqrt{17^2 - 8^2}$ ✓

$= \sqrt{225}$

$a = 15$ ✓

b



$7^2 = 49$

$2 \times b^2 = 49$

$b = \sqrt{24.5}$

$b = 4.95.$

(2 + 2 = 4 marks)

4 In each of the following, find the value of x correct to two decimal places.

a $\cos 70° = \frac{x}{3}$

$x = 3 \times \cos 70°$ ✓

$= 1.03$ ✓

b $\frac{x}{7.4} = \tan 28°$

$x = 7.4 \times \tan 28°$ ✓

$= 3.93$ ✓

c $\sin 34° = \frac{18}{x}$

$x = \frac{18}{\sin 34°}$ ✓

$= 32.19$ ✓

d $\frac{15.2}{x} = \cos 59.6°$

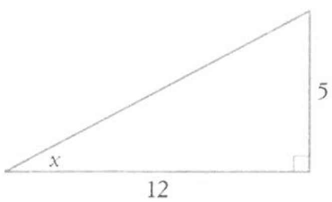
$x = 15.2 / \cos 59.6°$

$= 30.04$

(4 × 2 = 8 marks)

8 (4 marks each) Find the values of the pronumerals in each triangle. ~~Express angles in degrees and minutes~~. to 2dp

a

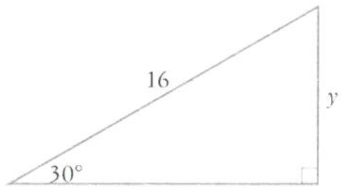


$\tan x = 5/12$ ✓

$x = \tan^{-1}(5/12)$ ✓

$= 22.62°$ ✓✓

b

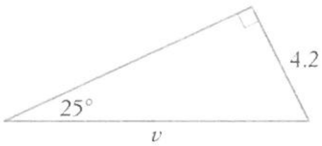


$\sin 30 = y/16$ ✓

$16 \times \sin 30 = y$ ✓

$y = 8$ ✓✓

c

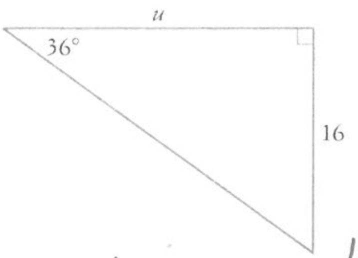


$\sin 25 = \frac{4.2}{v}$

$v = 4.2 / \sin 25$

$= 9.94$

d

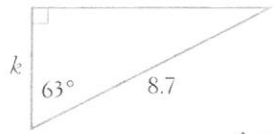


$\tan 36 = \frac{16}{u}$

$u = 16 / \tan 36$

$u = 22.02$

e

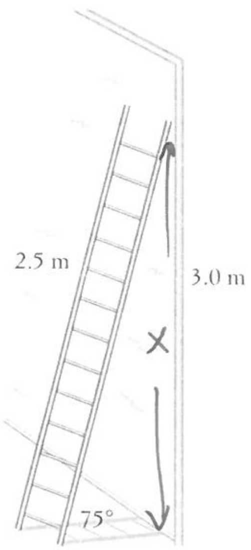


$\cos 63 = k / 8.7$

$8.7 \times \cos 63 = k$

$3.95 = k$

9 (5 marks) A 2.5 m long ladder leans against a 3.0 m high wall. If it rests against the (horizontal) ground at an angle of 75°, how far from the top of the wall will it reach? Use the space to the right of the diagram for working.



$\sin 75 = \frac{x}{2.5}$

$2.5 \times \sin 75 = x$

$x = 2.41$

DIST TO TOP $= 3 - 2.41$

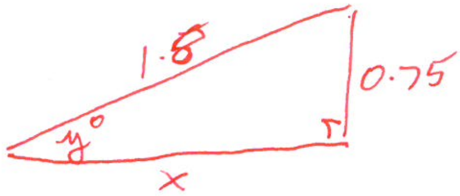
$= 0.59$ m

MUST HAVE UNITS
−1 IF NOT

10 A ramp is made from a rectangular piece of wood which is 1.8 m long. It connects a 75 cm high platform to the ground.

a (3 marks) How far away (horizontally) from the edge of the platform is the start of the ramp?

Draw a diagram



$x = \sqrt{1.8^2 - 0.75^2}$

$= \sqrt{2.6775}$

$x = 1.64$ m

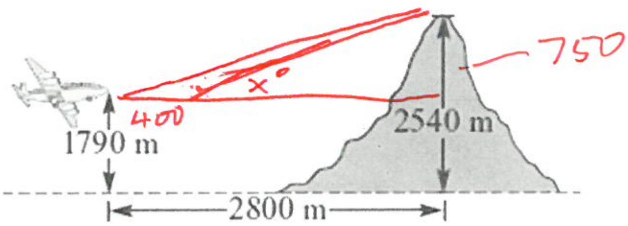
b (3 marks) At what angle is the ramp rising (to the horizontal)?

$\sin y = 0.75/1.8$

$y = \sin^{-1}(0.75/1.8)$

$y = 24.62°$

11 A pilot flying a plane at an altitude of 1790 m sees a mountain peak 2540 m high, 2800 m away from him in the horizontal direction.



a What is the difference in the vertical height between the plane and mountain peak?

$2540 - 1790 = 750$ m

b Find the direct distance between the plane and the mountain peak, correct to the nearest metre.

$\text{dist} = \sqrt{2800^2 + 750^2}$

$= \sqrt{8402500}$

$= 2898.71$ m

c If the pilot flies straight for another 400m find the angle the pilot would now have to climb at to just miss hitting the mountain peak(** round to the nearest degree). Would you round up or down and why?

HORIZONTAL DIST NOW 2400 m

$\tan x° = \frac{750}{2400}$

$x° = \tan^{-1}(750/2400)$

$= 17.35°$

WOULD ROUND UP SO HE GOES OVER MTN NOT INTO IT $\therefore$ 18°

(2 + 3 + 4 = 9 marks)